

THERMACART · DESIGN PRECIS

ThermaCart design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02: C5 conditioning step on the label, first candidate host partner, no ColdPod or ZeerBox dependency, low lip for wrong grades
0.5	2026-10-01	Amish Chadha	Constructable design (TCT-DDR-003) and prototype build plan (TCT-BLD-001); budget treated as a value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update from TCT-CAL-001 and TCT-DDR-001 (choices adopted for TRL 3 work, folding bail, cap stack, stock fin bar, verified PCM data, numbers, safety limits)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, grades, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

ThermaCart design precis

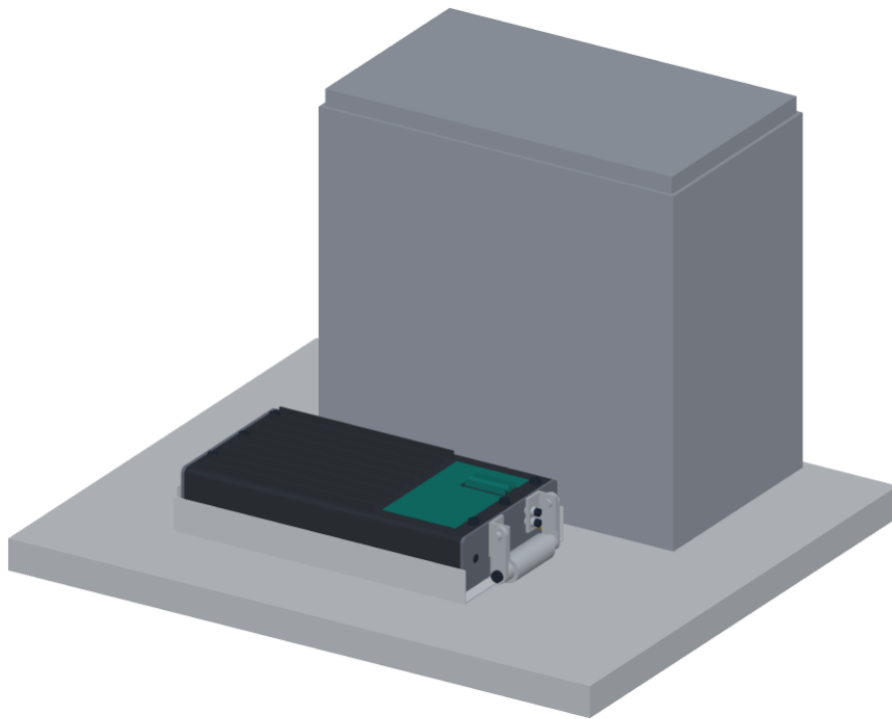
■ Summary

ThermaCart is a sealed aluminium cartridge, 323 x 152 x 64 mm, that holds 1.10 to 1.18 kg of phase-change material (PCM) in one of three standard grades: C5 (melts at 5 °C, cold chain), C25 (25 °C, comfort) and H70 (70 °C, hot holding). It fits a GN 1/3 gastronorm slot, carries by a folding bail, shows its state through a clear melt indicator and locks into an adapter frame whose key accepts only the right grade. The shell, fins and caps carry a matte black finish that improves heat exchange with the air. The TRL 3 calculations (TCT-CAL-001 v0.3) give 65, 61 and 73 Wh of usable storage per cartridge, 2.9 to 3.0 kg filled and about \$80 in parts. One requirement is not met on paper: the cold-chain hold time (R10, 2.4 h instead of 8 h, limited by heat transfer from the cooler air); the cost (R16) is over its value-engineering target. The design is constructable: every part can be cut, drilled, bent or bought and is fixed to the next (TCT-DDR-003, 2026-10-01), and the prototype build plan TCT-BLD-001 shows how to make it. The design choices below were decided by Amish on 2026-09-25 (TCT-DDR-001 and TCT-DDR-002).

ThermaCart

github.com/BoujeeEnjinia1701/thermacart

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: Table top, 25 L cooler for scale

Figure 1. A TC-L cartridge (C5 grade, matte black finish) in its adapter frame on a table, with a 25 L cooler for scale.

■ How it works

1. **Charge.** The cartridge is put where the right temperature already exists: a freezer for C5, a refrigerator or freezer for C25, and for H70 preferably a thermostatic heating pad, or an oven set to 85 °C (never above 90 °C). The PCM freezes (C5, C25) or melts (H70) and stores latent heat at a nearly constant temperature.
2. **Carry and swap.** The melt indicator shows the state. The owner lifts the bail, carries the cartridge to a cooler, food carrier, cabinet or heat exchanger and slides it nose first into an adapter frame.
3. **Hold.** The host box leaks heat in (cold grades) or out (H70). The PCM absorbs or releases it while staying near its melt point, so the box contents stay near that temperature for as long as the cartridge can move heat to or from them. In a produce cooler, the produce is laid directly on the cartridges, as with ice packs, so heat passes by contact as well as through the air (TCT-DDR-002, O5).
4. **Return.** The discharged cartridge goes back to the charging point. Nothing is consumed; the fill is sealed for life.

ThermaCart: energy per charge cycle, one TC-L H70 on a 60 W pad (TCT-CAL-001 estimates)

github.com/BoujeeEnjinia1701/thermacart

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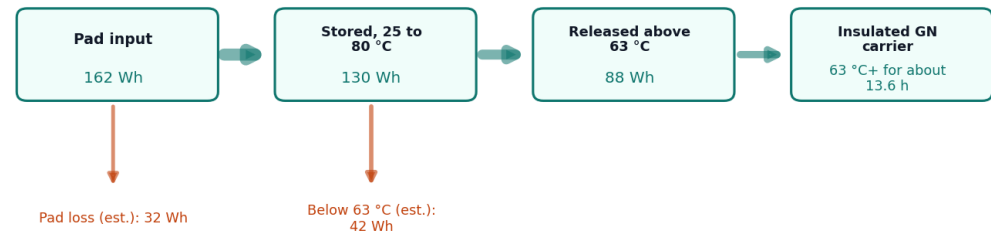


Figure 2. Energy per charge cycle for one H70 cartridge charged on a 60 W pad and used for hot holding (TCT-CAL-001, E4; estimates): 162 Wh in, 130 Wh stored between 25 and 80 °C, 88 Wh released above 63 °C and 42 Wh below it.

■ Standard grades

Table 1. Grades (decided by Amish, 2026-09-25)

Grade	Melt range	Fill (class)	Filled at	Usable storage per TC-L (TCT-CAL-001)	Typical hosts	Colour and key (Y of key centre)
C5	5 to 6 °C	Organic paraffin, Rubitherm RT 5 HC class	45 °C	65.4 Wh (1.104 kg)	Produce and dairy coolers; a ColdPod-type box after qualification	Blue, -43 mm
C25	22 to 26 °C (band 22 to 27 °C, TCT-DDR-002 09)	Organic paraffin, RT 25 HC class	65 °C	60.5 Wh (1.119 kg)	Electronics and battery cabinets, comfort cooling	Green, 0 mm
H70	69 to 71 °C	Organic paraffin, RT 70 HC class	90 °C	72.9 Wh (1.180 kg)	Insulated GN food carriers, keep-warm boxes	Red, +43 mm
W0 (option)	0 °C	Water with a nucleating agent	Not yet sized	Not yet calculated	Produce only; never for vaccines or medicines	White, -61 mm

Melt ranges, densities and maximum operating temperatures are from the Rubitherm datasheets ([RT 5 HC](#), [RT 25 HC](#), [RT 70 HC](#)). Each grade is filled at its maximum operating temperature (45, 65 and 90 °C), so the cartridge is never more pressurized in service than when it was sealed. Paraffins are used because they are chemically stable, non-corrosive to aluminium and cycle well ([PATH PCM study](#)). Salt hydrates would store more per litre but supercool, separate and corrode aluminium ([corrosion study](#)), so they are left for a later lined variant.

■ Main components

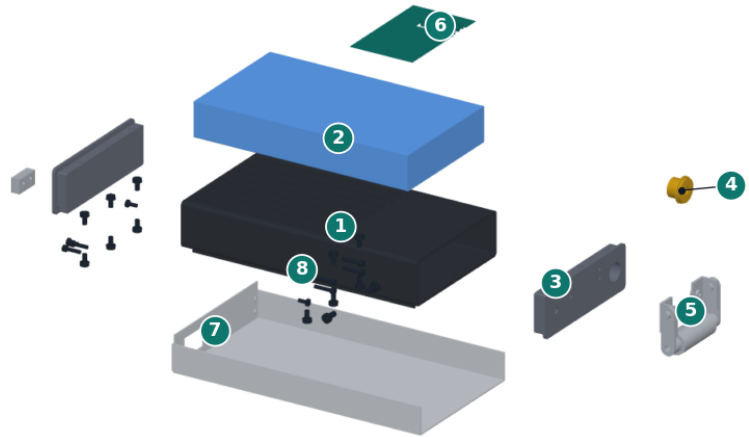
Table 2. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Choice for TRL 3	Notes
1	Cartridge shell	6063-T52 rectangular tube, 152.4 x 50.8 x 3.175 mm (6 x 2 x 1/8 in), 280 mm long, with nine 6.35 x 1.59 mm (1/4 x 1/16 in) fin bars bonded on edge on each broad face; outer surfaces etch-primed and painted matte black with high-temperature paint	Stock sections, no welding; fins add 44 % to the tube's outer area; the finish raises emissivity from about 0.1 to 0.9 (TCT-DDR-002, O8)
2	PCM fill	1.10 to 1.18 kg of the grade's PCM, filled as liquid to 90 % of the 1.655 L inner volume	See Table 1 and Safety
3	End caps (2)	Stack of 3.175 mm flange, 3 mm gland spacer and 9.525 mm plug in 6061 plate, bonded face to face and screwed; spliced FKM 2.5 mm cord O-ring in the corner gland; six radial M5 button-head screws on bonded seals through the tube wall into the plug	Cut and filed, no milling; FKM (or NBR) resists paraffin, EPDM does not
4	Fill port plug	G 1/2 anodised aluminium plug with a bonded FKM seal in the handle-end cap, clear of the O-ring and the bail	Fill, drain and recycling
5	Handle and keyed nose	Folding bail (16 mm rod, silicone grip, two 16 x 4 mm arms on 5 mm pins) on two 20 x 20 x 3 mm lug angles bolted to the cap through 3 mm phenolic thermal-break washers, stowed against the end face; key tab screwed to the nose at the grade position	41 mm finger gap when raised; key positions in Table 1
6	Grade label and melt indicator	Colour-coded polyester label; a clear polycarbonate tube holding the same PCM, bonded on the top between two guards; H70 label carries the oven handling rule	Paraffin is opaque white when solid and clear when liquid
7	Cabinet adapter frame	Bent 1.5 mm 5052 aluminium sheet, 294 x 160 mm, with side walls riveted to a keyed end stop; the key passes through the stop's slot when seated	One per host slot; host designs copy its key slot
8	Hardware	Twelve M5 button-head screws with bonded seals, nine M4 screws and bolts, two M6 rod screws (A2), epoxy rated 120 °C or more, thread sealant	Per cartridge

ThermaCart: exploded viewgithub.com/BoujeeEnjinia1701/thermacart

CONCEPT, NOT FOR FABRICATION

- 1 Cartridge shell, finned aluminium tube, matte black
- 2 PCM fill, C5 grade (5 °C)
- 3 End caps with O-ring seals (2)
- 4 Fill port plug and seal
- 5 Handle (folding bail) and keyed nose
- 6 Grade label, melt indicator and guards
- 7 Cabinet adapter frame
- 8 Screws, bolts and seals



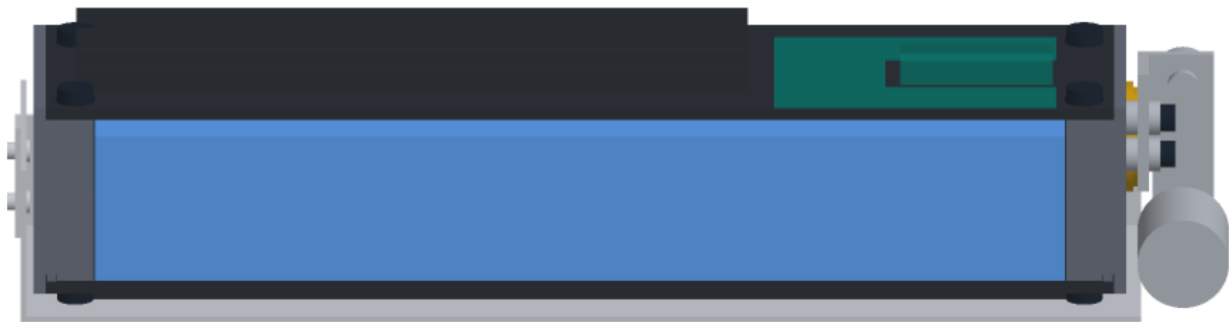
Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view. 1 shell, 2 PCM fill, 3 end caps, 4 fill port plug, 5 handle and keyed nose, 6 label and indicator, 7 adapter frame, 8 cap screws.

ThermaCart: cutaway

github.com/BoujeeEnjinia1701/thermacart

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 4. Section along the cartridge. The PCM (blue) fills 90 % of the inner height when liquid; the rest is ullage. The cap stacks, fins and label band are visible. The general arrangement is drawing TCT-DWG-001.

■ Key numbers

All values are from TCT-CAL-001 v0.2 (dark finish as the design case) and the parametric model `cad/src/model.py`.

Table 3. Key numbers

Quantity	Value	Requirement
Overall size	323.4 x 152.4 x 63.5 mm, bail stowed	R2 met (GN 1/3: 325 x 176 mm, 65 mm)
Inner volume	1.655 L (146.05 x 44.45 x 254.9 mm)	
PCM mass	C5 1.104 kg, C25 1.119 kg, H70 1.180 kg	
Usable storage within ± 3 K	C5 65.4 Wh, C25 60.5 Wh, H70 72.9 Wh	R3 met (C25 thin)
Empty cartridge	1.80 kg	
Filled mass	2.90 to 2.98 kg	R4 met (thin)
PCM fraction	38.1 to 39.6 %	R5 met (thin; target 38 % for the stock tube)
Pressure	0.00 bar gauge at the fill temperature, -0.68 bar when frozen, +0.29 bar at 20 K over the limit; wall stress 56 MPa against 73 MPa allowable	R8 met
Cold hold, two C5 in a 25 L cooler at 32 °C	2.4 h (1.9 h bare); stored energy alone would last 9.7 h; produce contact not credited	R10 not met
Hot hold, one H70 in a GN carrier at 25 °C	13.6 h at 63 °C or more (7.2 h bare)	R11 met
Recharge	C5 in a freezer 4.2 h; C25 in a refrigerator 4.6 h; H70 in an 85 °C oven 5.3 h, on a 60 W pad 2.7 to 10.3 h	R12 at risk (pad)
Bail temperature, H70	39.1 °C on its thermal break; 85 °C for about 10 min after an oven	R14 met with the label rule
Parts cost	About \$80 per cartridge, \$17 per frame (indicative)	R16 over its value-engineering target by \$30 per cartridge; one cartridge and one frame \$97 against the \$80 value-engineering target

■ Key design choices (decided by Amish, 2026-09-25)

Each choice below was decided by Amish on 2026-09-25, going with the recommendation (TCT-DDR-001, items D1 to D8 and T1 to T4, and TCT-DDR-002, items O4 to O9).

- Envelope: GN 1/3 footprint from a stock tube (D1).** A 6 x 2 in stock aluminium tube cut to fit a GN 1/3 slot. The WHO PQS 0.6 L pack envelope, 190 x 120 x 34 mm (WHO PQS E005), is kept as a later small size (TC-S), and a custom finned extrusion only if volumes justify a die.
- Three paraffin grades, water as an option, C5 first (D2).** See Table 1.
- Mechanical grade keying and a passive indicator (D4).** A nose tab whose position depends on the grade and a matching slot in each frame, so a frozen water or hot H70 cartridge cannot be put into a cold-chain box; the key table is published as part of the open standard. A clear indicator holding the same PCM shows solid or liquid without electronics.
- The 2 mm wall is not used (D5).** It adds 3.5 % volume and would yield under the vacuum of a frozen cartridge. C5 meets R3 on the 3.175 mm tube.

5. **Fill hot, seal for life (D8).** Filling as liquid at the grade's maximum operating temperature fixes the largest PCM volume at sealing, so the cartridge only sees vacuum below that temperature and a small positive pressure if the limit is exceeded.
6. **Aluminium shell for paraffins, no HDPE (D8).** Paraffin can swell and permeate polyethylene ([compatibility study](#)); aluminium is compatible and conducts heat to the fins. A lined or HDPE variant is kept for salt hydrates and water.
7. **Folding bail, cap stack and stock fin bar (T1 to T3).** Refinements that keep the cartridge inside the GN 1/3 envelope, give room for fingers and avoid milling.
8. **Matte black finish (O8).** The shell, fins and caps are etch-primed and painted with a high-temperature paint. It raises the air-side coupling from 0.50 to 1.11 W/K per cartridge and helps R10, R11 and R12 for about \$5 and 15 g. The bail stays bare aluminium on its thermal break.
9. **Produce on the cartridges (O5).** User guidance for coolers is to lay produce directly on the C5 cartridges. The contact effect is not credited on paper and waits for a TRL 4 measurement, which is on hold; R10 keeps its 8 h target.
10. **H70 handling rule and pad charging (O7).** The H70 label says: from an oven, use oven gloves or wait 12 min before carrying by the bail. Pad charging is the preferred method.
11. **Requirement changes (O4, O9).** R5 is 38 % for the stock-tube build, with 50 % kept for a later extrusion; the C25 band is 22 to 27 °C.
12. **Cost (O6).** Supplier quotes come first; quotes are a purchasing step and wait with TRL 4, which is on hold. `budget_usd` (\$80) is a hypothetical value-engineering target, not a limit (Amish, 2026-10-01).

■ Safety

Safety: Paraffin fills are combustible. Never charge an H70 cartridge on an open flame, a gas ring or a hot plate. Set an oven to 85 °C and check it with an oven thermometer; never let an H70 cartridge exceed 90 °C, the maximum operating temperature of RT 70 HC. If a cartridge leaks, stop using it; spilled liquid paraffin is slippery and burns.

Safety: Keep C5 cartridges below 45 °C and C25 cartridges below 65 °C, their fill temperatures and datasheet limits; do not leave them in a closed car in the sun.

Safety: H70 cartridges reach 70 to 85 °C and can burn skin. Carry them only by the bail. Straight out of an oven the bail and its grip are at oven temperature for about 12 min: use oven gloves or wait 12 min. Prefer pad charging. The H70 label carries this rule and a hot-surface warning.

Safety: Do not fill, refill or open a cartridge that is warm or under pressure. Fill only as described in the fill recipe, with 10 % ullage at the grade's fill temperature. A frozen cartridge holds a partial vacuum of about 0.7 bar; its broad faces pull in by about 0.8 mm, which is expected.

Safety: A C5 cartridge charged in a freezer is below 0 °C when it comes out. Condition it until the melt indicator shows the first liquid before using it next to anything that must not freeze; the conditioning time is printed on the C5 label as a required step (decided by Amish, 2026-10-02). ThermaCart is not qualified for vaccines or medicines, and the W0 water grade must never be used with them.

- Aluminium edges and fins are sharp after cutting: deburr all edges.

- Use a high-temperature paint rated well above 90 °C for the dark finish, paint in a ventilated space, and keep paint off the fill port thread and seat, the O-ring glands, the bores and the indicator.
- A 3 kg cartridge dropped on a foot can injure: keep cartridges low in racks. A drop of a molten cartridge may stress the seals (TCT-CAL-001, C7).

■ Open questions after TRL 3

- R10: the dark finish and produce contact are decided; whether contact closes the gap to 8 h needs a measurement at TRL 4, which is on hold.
- R16: \$80 per cartridge, \$30 over its \$50 value-engineering target; supplier quotes wait with TRL 4. Savings worth trying are listed in the design decisions register (TCT-DEC-001).
- C5 charging (O3, decided by Amish, 2026-10-02): a domestic freezer followed by the conditioning step, about 4.2 h against 31 to 51 h in a refrigerator, with the conditioning time printed on the C5 label as a required step.
- Seal performance over cycles and in a drop (R6), cycle life (R7) and the pad charge time (R12) need tests, which belong to TRL 4 and are on hold.
- First host (O1, decided by Amish, 2026-10-02): a produce cooler user reached through a postharvest extension partner; the first candidate to approach is the UC Davis Postharvest Technology Center, asked to introduce a grower group or market cooler operator that already uses passive or ice-based cooling. Nothing is agreed. Neither ColdPod nor ZeerBox declares ThermaCart as a dependency now (O2).
- Wrong grade (R9, decided by Amish, 2026-10-02): a low lip at the frame's open end, with the stowed grip moved up, so that the frame physically rejects a wrong grade; it is still to be designed and modelled.

Prototype build plan: [TCT-BLD-001](#). Design decisions register: [TCT-DEC-001](#). Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [TCT-DWG-001](#).